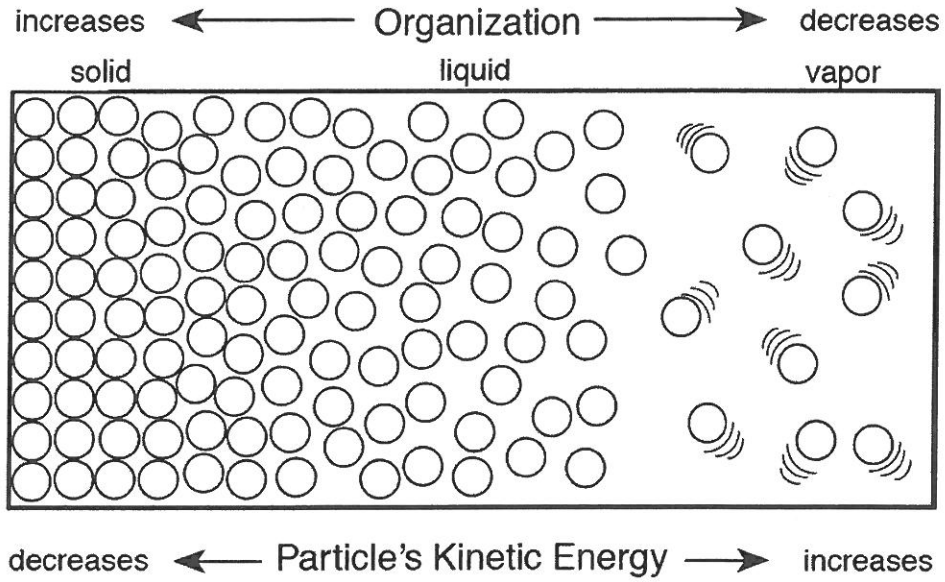
YEAR 8 SCIENCE
EXTENDED ANSWER QUESTION: PHASES CHANGES

Look at the diagram below.



PHASE CHANGES ON A MOLECULAR LEVEL

In three paragraphs (maximum), explain what is being shown in this diagram.

Your first paragraph should be a short introduction. The second and third should explain the processes being illustrated with a concluding sentence (or two) at the end of the third paragraph. Remember to use 'linking sentences' at the end of paragraphs one and two.

List the scientific terms or words you should use in the section on the answer page and plan your answer in the space below before starting your essay.

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NAME: .....

OPEN ENDED QUESTIONS

Question: Explain what is being shown in the diagram

Key words/terms that need to be included:

states of matter
organisation
kinetic energy
motion
spacing
particles
lattice
random
straight line motion
heating
increasing/decreasing
bonds

The diagram uses circles to represent molecules in all three states of matter.

In solids particles are held in a highly organised lattice structure. The particles are only able to vibrate around a fixed position. As kinetic energy increases from left to right the particles become less well organised. On the right they are completely disordered and moving in rapid random, straight line motion.

Kinetic energy on the diagram is a measure of the amount of movement energy the particles have. As the movement energy increases from left to right the particles move more rapidly and are able to overcome the bonds between them. There is an inverse relationship between the amount of organisation and kinetic energy. In other words, as kinetic energy increases, organisation decreases.